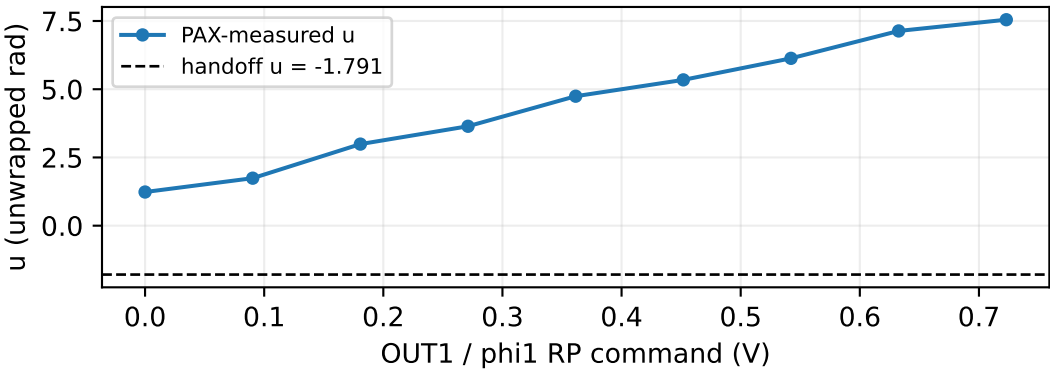
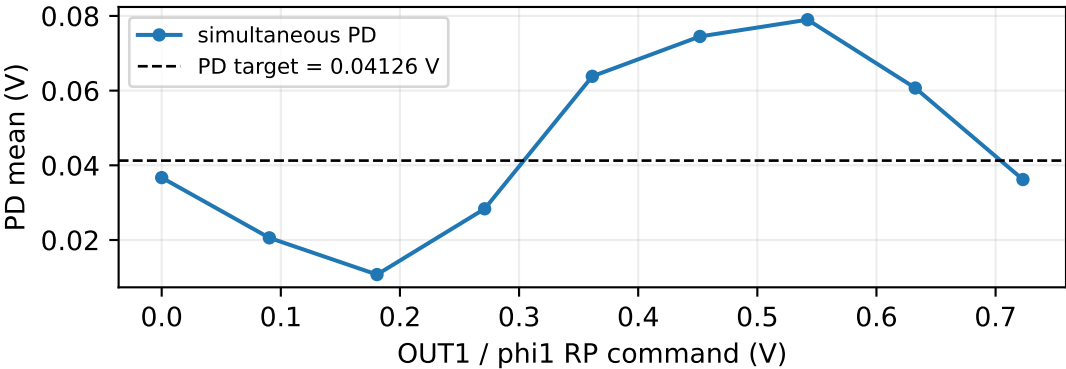


Phi1 PD FPGA lock — 014959_phi1-pd-hybrid-test_phi1-pd-hybrid-1

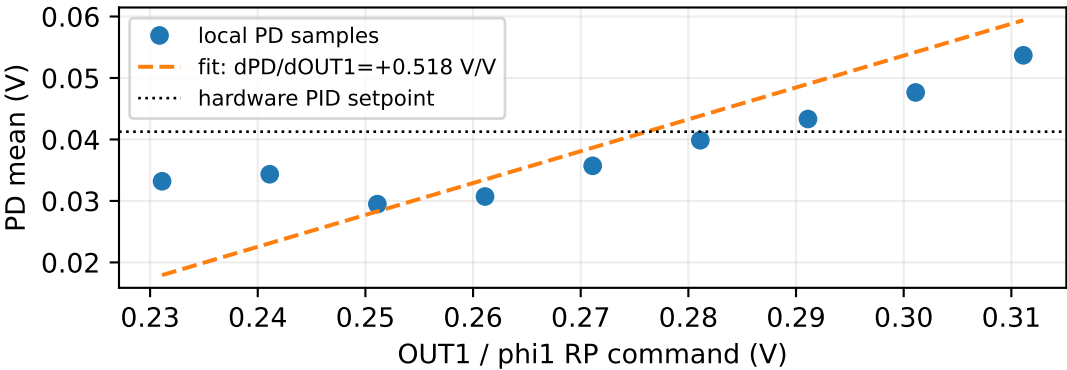
PAX-assisted ϕ_1 rough map



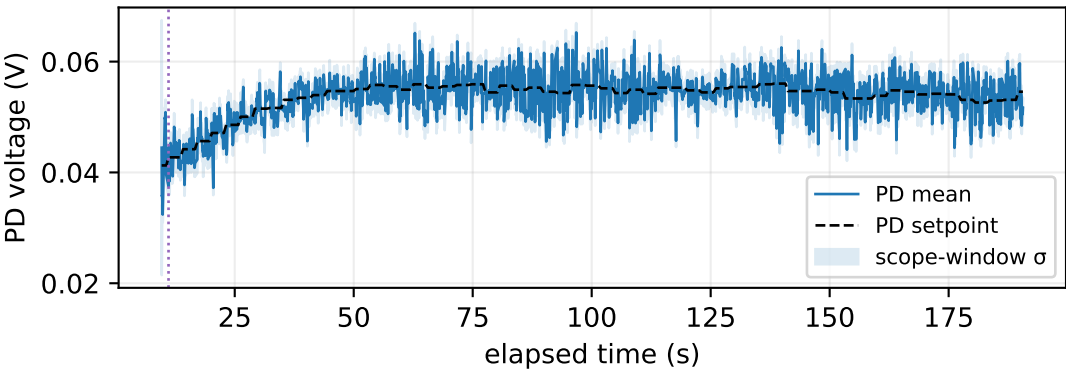
PD value associated with rough u map



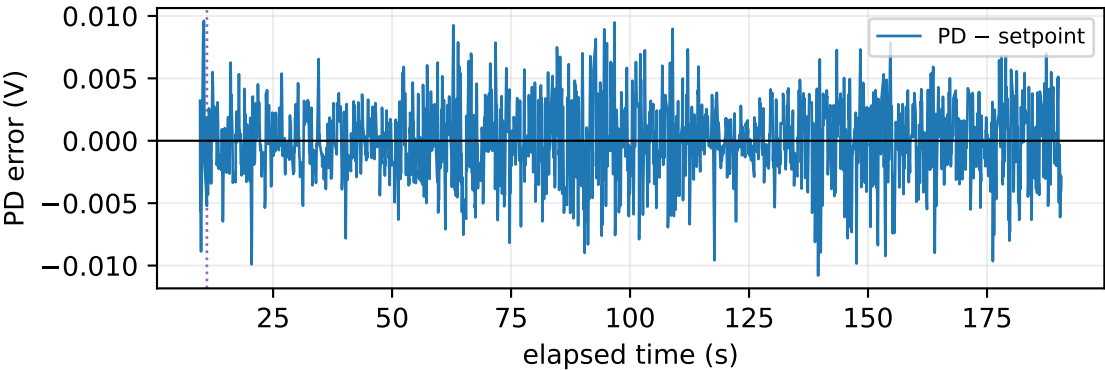
Automatic local PD calibration



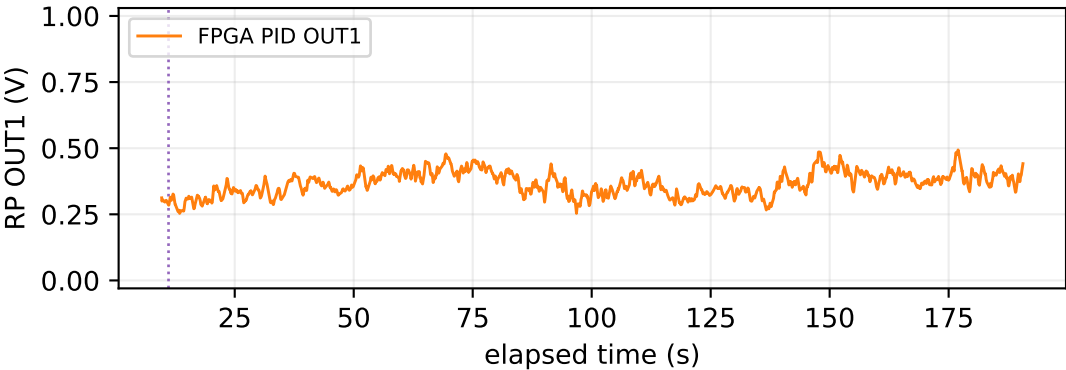
FPGA PD lock



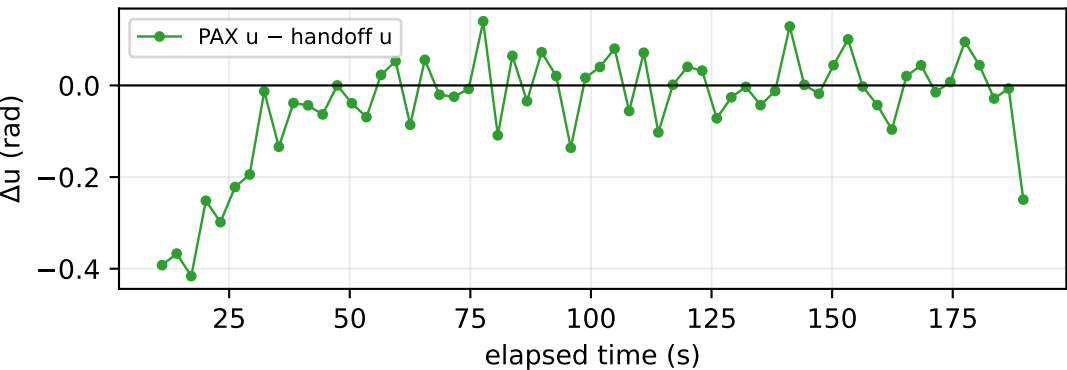
PD locking error



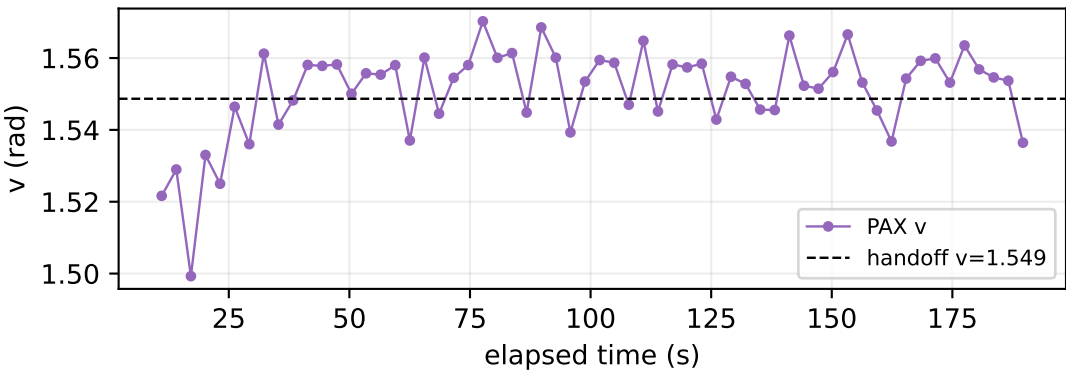
Continuous FPGA actuator output



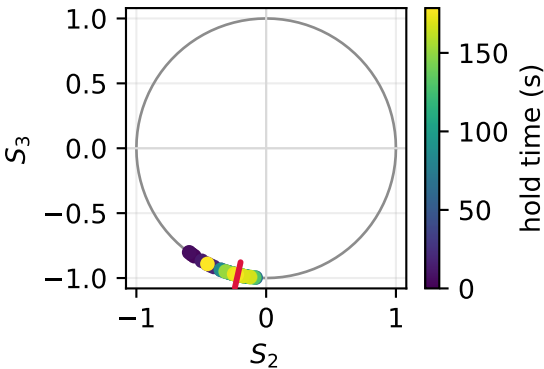
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



PAX handoff target $u = -1.7914$ rad | initial PD setpoint = 0.04126 V | local slope = +0.5182 V/V
FPGA PID: P = -0.425, I = -5 Hz | duration = 181.0 s | median |PD error| = 0.00184 V (late 0.00196 V)
OUT1 = 0.253...0.493 V | PD setpoint = 0.04126...0.05603 V | PAX median $|\Delta u|$ = 0.044 rad (60 snapshots)